



TEMASEK JUNIOR COLLEGE

Preliminary Examinations 2013

BIOLOGY
Higher 2
Paper 2 Core Paper

9648/02
Monday, 16 September
2 hours

Candidate Name: _____

Civics Group: _____

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Part I and Part II)

Answer **all** questions

Section B

Answer any **one** question

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [] at the end of each question or part question.

For examiner's use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 / 9	
TOTAL	

This document consists of **25** printed pages and **1** blank page.

[Turn over

Section A [Part 1]For
Examiner's
UseAnswer **all** the questions in this section.

- 1** Oxygen is an essential molecule for eukaryotic organisms as it is needed for the generation of ATP for cellular processes. In mammals, oxygen is carried by haemoglobin. Haemoglobin consists of 4 globin subunits, and each subunit has a prosthetic group called the heme group, where an iron ion binds to an oxygen molecule.

- (a)** Name the precise type of bond between the monomers in haemoglobin and describe how it is formed. Use an annotated diagram in the space below to illustrate your answer.

Name of bond:

Formation:

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.....

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Diagram:

[4]

[Turn over]

Fig. 1.1 shows the structure of one of the haemoglobin subunits with the prosthetic group.

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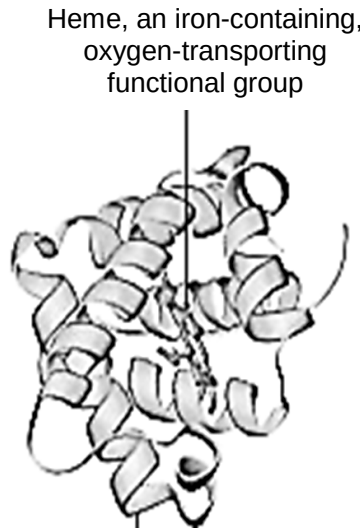


Fig. 1.1

(b) Describe the level of protein structure seen in each globin subunit.

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..... [3]

Haemoglobin has the ability to bind to other molecules through the same heme group. Fig. 1.2 shows the affinities of different types of haemoglobin for molecule X at different concentrations of the molecule (reflected by its partial pressure).

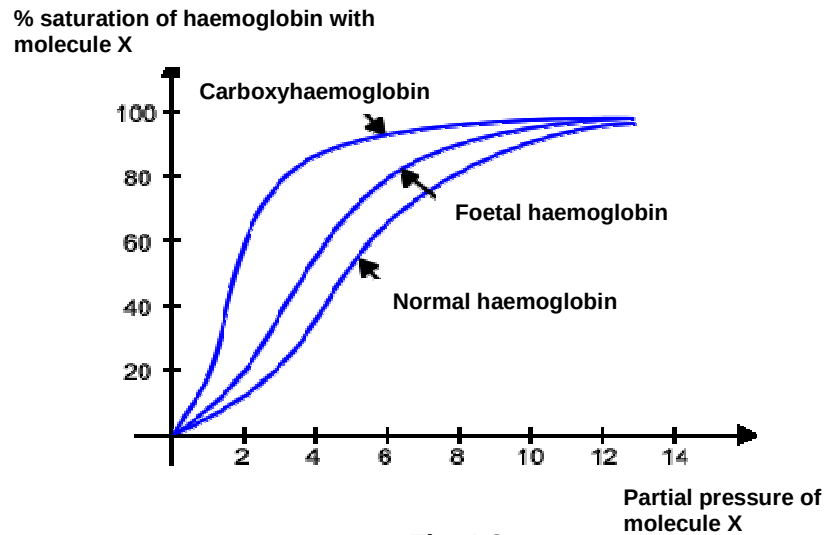


Fig. 1.2

Another example of a molecule that binds to haemoglobin is carbon monoxide (CO). It is commonly found in cigarette smoke, smoke from car exhaust pipes etc., and when bound to haemoglobin, it forms a molecule called carboxyhaemoglobin. Small amounts of CO can dramatically reduce haemoglobin's ability to transport oxygen. When inspired air contains as low as 0.02% CO, headache and nausea can set in. 0.1% CO, the person will become unconscious.

- (c) (i) Suggest whether haemoglobin has a higher or lower affinity for CO as compared to oxygen.

..... [1]

- (ii) Explain the physiological effects of CO poisoning in a human.

..... [2]

(iii) Suggest how CO poisoning can be treated.

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..... [1]

[Total: 11]

[Turn over

- 2 Fig. 2.1 shows a cell undergoing gametogenesis in the plant *Lilium grandiflorum*.

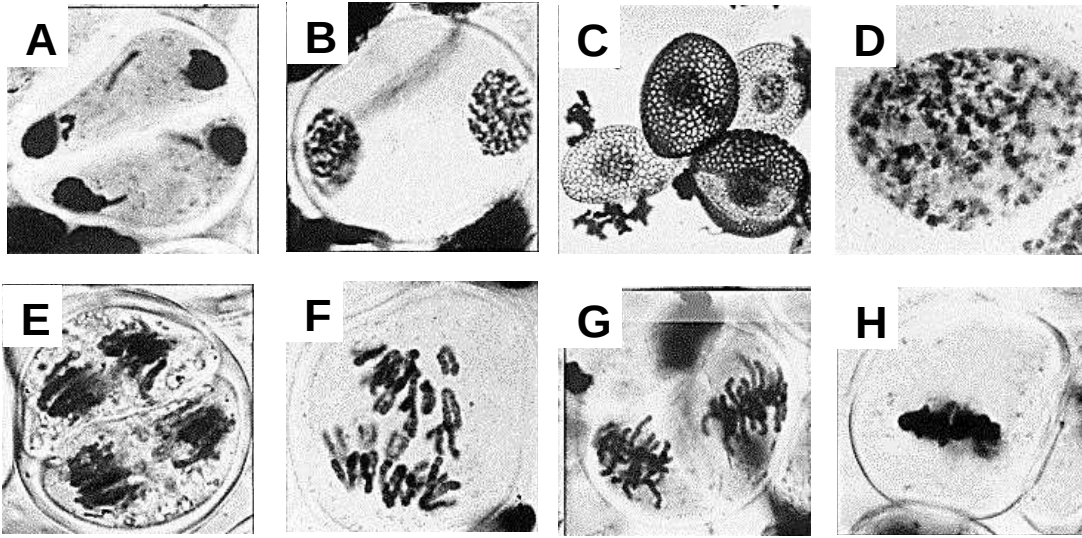


Fig. 2.1

- (a) Arrange the letters **A** to **H** in Fig. 2.1 to show the correct chronological sequence of events for this type of nuclear division.

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[2]

- (b) Describe what occurs inside the cell during
(i) stage **A**.

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.....

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..... [2]

- (ii) stage **F**.

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..... [2]

(c) Explain the significance of this nuclear division during gametogenesis.

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..... [2]

During the nuclear division illustrated in Fig. 2.1, X and Y are structures found in the cell at a particular stage, shown in Fig. 2.2.



Fig. 2.2

(d) Describe **one** way that X and Y are

(i) genetically similar.

.....

..... [1]

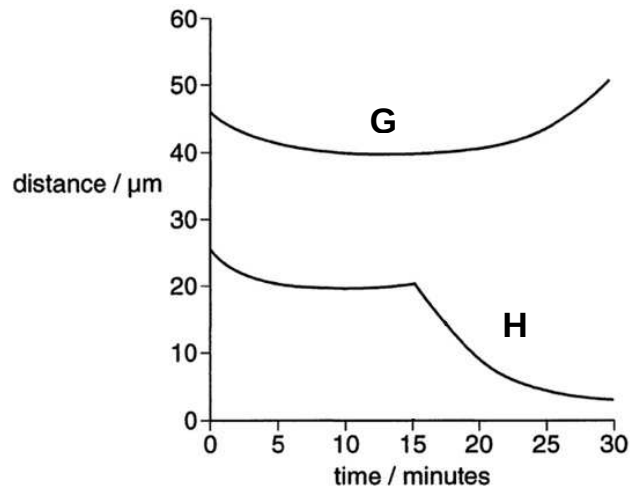
(ii) genetically different.

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..... [1]

This type of nuclear division was also studied using *Caenorhabditis elegans*, a transparent nematode (roundworm) that is commonly used as a model organism for scientific research. Researchers have found that mutation of the β -tubulin gene *tba-2* disrupts nuclear division, resulting in abnormal and non-viable embryos.

Fig. 2.3 shows the distance between poles of the spindle (graph **G**) and distance between centromere and poles of the spindle (graph **H**) for wild type *Caenorhabditis elegans* during meiosis I.



[1]

Fig. 2.3

- (e) On Fig. 2.3, sketch the graph for distance between centromere and poles of the spindle for *Caenorhabditis elegans* with mutated *tba-2* gene. Explain your answer.

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..... [2]

[Total: 13]

- 3 In the bacteria species, *Lactobacillus*, the production of the amino acid arginine is catalyzed by five specific enzymes encoded by specific genes *argC2*, *argJ*, *argB*, *argD* and *argF* as shown in Fig. 3.1. The *arg* operon is transcriptionally regulated by 2 repressor proteins, R1 and R2, encoded by the *argR1* and *argR2* genes, which are located far away from the *arg* operon.



Fig 3.1

When levels of arginine are high, six arginine molecules bind to three R1 and three R2 molecules to form a repressor complex. The repressor complex then binds to the operator region.

- (a) (i) With reference to Fig 3.1, identify the regulatory and structural genes.

Regulatory genes:

Structural genes: [1]

- (ii) Distinguish between regulatory gene and structural gene.

.....

 [2]

- (b) Explain why it is useful for a bacterial cell to decrease expression of the *arg* genes when arginine is present.

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 [2]

Table 3.1 below indicates the activity levels of the functional enzymes C2, J, B, D and F in wild type bacterial cells in the presence and absence of arginine (Arg).

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Table 3.1

Enzymes	Activity level of enzymes/units	
	Arg absent	Arg present
C2	500	0
J	500	0
B	500	0
D	500	0
F	500	0

Three bacterial mutants were discovered. Each mutant is the result of a single base substitution in a component of the *arg* operon. The activity level of functional enzymes C2, J, B, D and F in the mutant bacterial cells is shown in Table 3.2.

Table 3.2

Enzymes	Activity level of enzymes/units					
	Mutant 1		Mutant 2		Mutant 3	
	Arg absent	Arg present	Arg absent	Arg present	Arg absent	Arg present
C2	500	500	500	0	0	0
J	500	500	0	0	0	0
B	500	500	500	0	0	0
D	500	500	500	0	0	0
F	500	500	500	0	0	0

- (c) (i) State the mutant with the phenotype that is consistent with a loss-of-function mutation in the *argR* genes. Explain your choice using the data from Table 3.2.

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- (ii) Suggest the components of the *arg* operon that is mutated to give rise to the other two mutants that are **not** mentioned in your answer in (c) (i).

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..... [2]

[Total: 11]

- 4 Antigens are found on the human red blood cell and are used to determine the human ABO blood group system. Type and presence of antigens are determined by three alleles, I^A , I^B and I^O . Alleles I^A and I^B are co-dominant and both alleles are dominant to allele I^O .

(a) Define the term co-dominance using the example of the ABO blood group.

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..... [2]

ABO haemolytic disease of the newborn refers to blood disorders that arise when the blood groups of the mother and newborn child do not match. It usually happens during a mother's second or subsequent pregnancy.

Antibodies from the mother can pass through the placenta and bind to the child's red blood cells. This leads to cell lysis, resulting in anaemia. Table 4.1 shows the three different situations where blood group incompatibility can arise between a mother and her child.

Table 4.1

	Blood group		
	Situation 1	Situation 2	Situation 3
Mother	O	A	B
Child	A or B	B	A

A man with blood type A married a woman with blood type O.

(b) Use genetic diagrams to show the genotypes of the children that the couple can have.

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[4]

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- (c) (i) With reference to Table 4.1 and your answer to (b), calculate the probability that the second newborn will suffer from ABO haemolytic disease. Show your working.

[1]

- (ii) Suggest a possible treatment if the newborn suffers from severe anaemia due to ABO haemolytic disease.

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..... [1]

The ABO blood group system is a type of discontinuous variation, while in some phenotypic characteristics such as skin colour, they exhibit continuous variation.

- (d) Distinguish between the **two** types of variation in the ABO blood group system and skin colour.

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..... [3]

[Total: 11]



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BIOLOGY
Higher 2
Paper 2 Core Paper
Section A [Part 2]

For examiner's use	
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Candidate Name:

Civics Group:

- 5** Fructose 1, 6-bisphosphatase is a crucial enzyme involved in the process of regenerating ribulose biphosphate. Studies using etiolated spinach seedlings show residual fructose 1, 6-bisphosphatase activity, which sharply rises under illumination.

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Fig. 5.1 shows a section of the organelle where fructose 1, 6-bisphosphatase can be obtained.

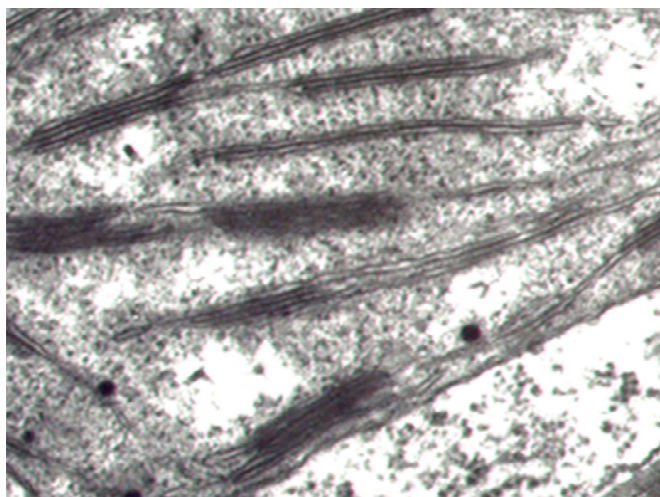


Fig. 5.1

- (a) (i)** State the precise location of fructose 1, 6-bisphosphatase activity.

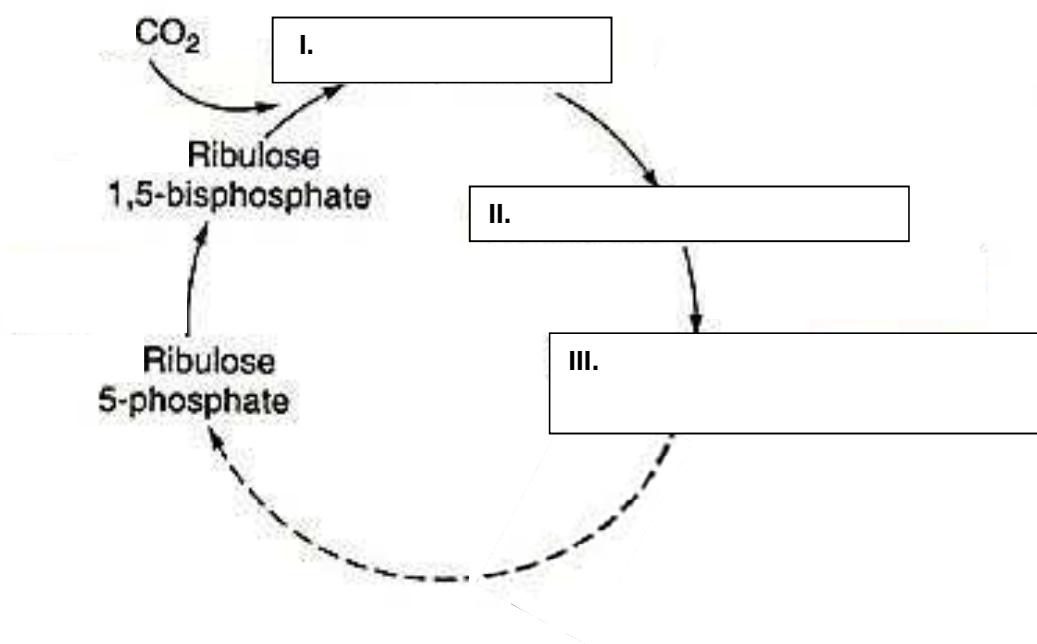
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- (ii)** Annotate in Fig. 5.1 your answer for **(a) (i)**.

[1]

[Turn over

(b) The following cycle also occurs in the same organelle.



(i) Fill in the blanks to complete the cycle.

[3]

(ii) Draw the structure of the storage form of the sugar produced from the cycle in the space below. Indicate the different types of bonds present.

[2]

[Turn over

(iii) Explain **two** key functions of the cycle, other than the synthesis of sugar.

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..... [2]

Cycloheximide is an inhibitor of protein biosynthesis in eukaryotic organisms. The rise of fructose 1, 6-bisphosphatase activity under illumination has been found to be strongly inhibited by cycloheximide, but not by lincocin. Lincocin is a 70S ribosome inhibitor.

(c) Explain if the gene coding for fructose 1, 6-bisphosphatase is found in the nucleus or the chloroplast.

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..... [2]

[Total: 10]

- 6 Fig. 6.1 shows the structure of a glucagon molecule, which is made up of 29 amino acid residues.

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Use



Fig. 6.1

- (a) State the target cell of the hormone glucagon.

..... [1]

- (b) Explain how glucagon helps to maintain blood glucose level.

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..... [3]

In recent studies, it has been found that glucagon-like peptides such as GLP-1 also help to control blood glucose level in normal individuals. GLP-1 is secreted by intestinal cells when stimulated by the presence of carbohydrate, protein and lipid. Once in the blood circulation, GLP-1 serves to stimulate insulin production and suppress glucagon secretion, thereby ensuring that the blood glucose level is normal. GLP-1 has a half-life of less than 2 minutes and will be rapidly degraded by the enzyme dipeptidyl peptidase-4.

GLP-1 has been found to bind to G-protein coupled receptors. This stimulates a cell signalling pathway which leads to the transcription of *IRS-2* gene, resulting in increased cell proliferation in beta cells in the pancreas (Brubaker & Drucker, 2004). Fig. 6.2 illustrates one such pathway in beta cells.

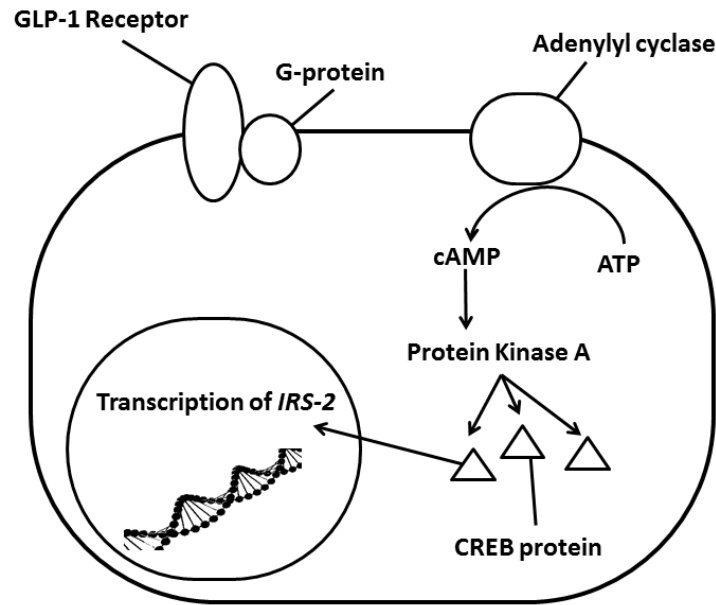


Fig. 6.2

- (c) With reference to Fig. 6.2, describe how the binding of GLP-1 to its receptor results in cell proliferation.

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[4]

A new drug has been designed for the treatment of diabetes mellitus type II. The drug targets the action of dipeptidyl peptidase-4.

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(d) Using the above information, explain the mechanism of action of this drug.

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..... [2]

(e) Describe **two** advantages of having a cell signalling system.

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..... [2]

[Total: 12]

- 7 Heterozygote advantage was first confirmed based on an experiment using *Drosophila melanogaster*, a fruit fly, as the model organism. In a classic study, Kalmus discovered a mutant allele of an autosomal gene, which will give rise to ebony body colour (Fig. 7.1). When a fly inherited two copies of the mutation, it expressed the dark ebony colour and was also particularly weak. However, when only one copy was inherited, the same mutation resulted in improved viability for the individuals.

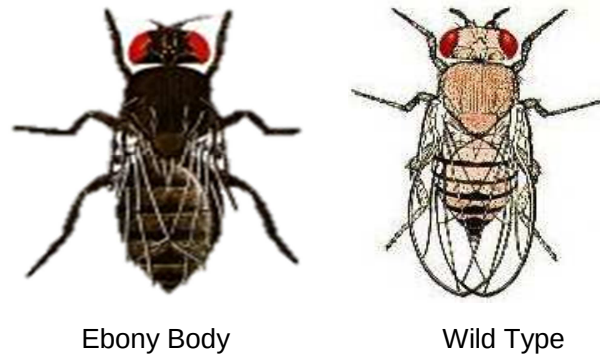


Fig. 7.1

In this study, Kalmus introduced flies with the ebony mutation to a wild-type population. The mutant allele persisted through many generations of flies, at genotype frequencies that varied from 8% to 30%. The mutant allele was more prevalent when flies were bred at lower temperatures and drier environments.

- (a) Explain how heterozygote advantage in *Drosophila* can ensure that variation is preserved in a population undergoing natural selection.

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..... [4]

- (b) Outline the effects on the allele frequencies in the *Drosophila* population when the climate changes.

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..... [2]

- (c) Suggest why *Drosophila melanogaster* has been a model organism for this study.

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..... [2]

Investigation of the evolutionary origin of avian feathers has been constrained by the lack of any known ancestral feather morphologies or structural antecedents. Feathers first appeared in the fossil record in *Archaeopteryx* and some scientists have proposed a model to explain the evolutionary diversification of avian feathers. Fig. 7.2 shows the different stages of feather development, where Stage I illustrates the earliest form and Stage V is the most advanced form. The different feather morphologies are still present in extant species.

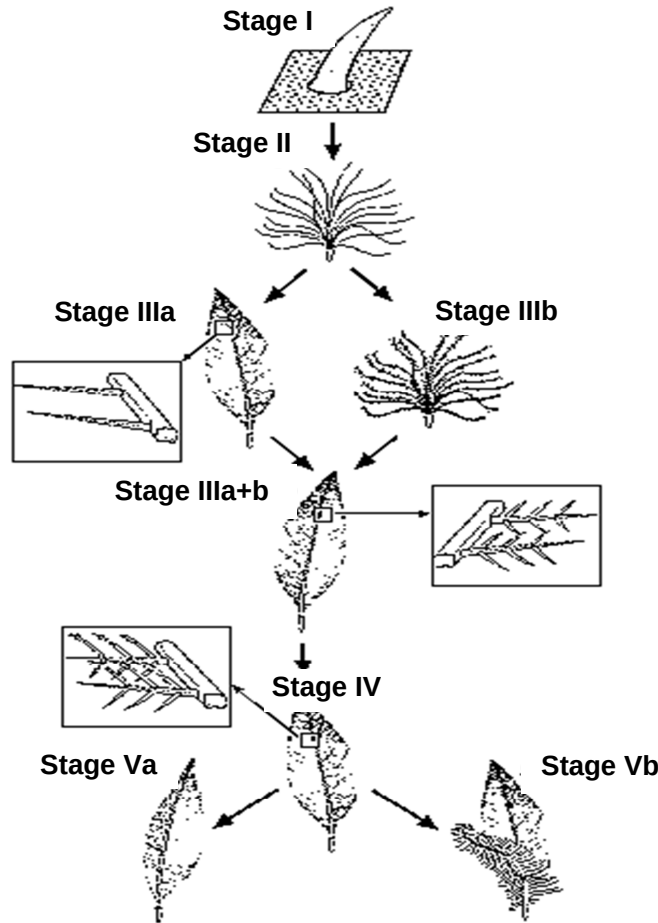


Fig. 7.2

(d) Explain how the proposed model can be explained through genetic drift.

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[3]

Homology among structures can be determined through anatomy, embryology and molecular studies.

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(e) Suggest why homology cannot be determined by functional criteria.

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..... [1]

[Total: 12]

SECTION B**Free response question****Answer one question.**

Write your answers to this question on the separate answer paper provided.

Your answers

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 (a) Describe DNA replication of the leading and lagging strands. [7]

(b) Discuss the roles of mRNA, tRNA and rRNA in protein synthesis. [5]

(c) Using a named example, discuss how gene amplification regulates expression of gene and its significance in regulation of gene expression. [8]

[Total: 20]

OR

9 (a) Explain what is meant by an endocrine gland, with reference to the islets of Langerhans in the pancreas. [7]

(b) Describe how an action potential is transmitted along a myelinated neurone. [7]

(c) Explain how a nerve impulse is passed across a cholinergic synapse. [6]

[Total: 20]

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